Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3291-WE01

Title: (BEng/MEng)

Thermodynamics and Fluid Mechanics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

(a) An axial turbine stage design is being considered. Starting from a sketch of the stage cascade view and Euler equation, show that if the flow through this stage has constant axial velocity, the stage loading coefficient ψ and the flow coefficient ϕ are related by:

$$\psi = |1 - \phi\,(\tan\alpha_2 - \tan\beta_3)|,$$

where α_2 is the absolute flow angle upstream of the rotor and β_3 is the relative flow angle downstream of the rotor.

[25%]

(b) The stage is now designed to be a zero reaction (impulse) stage where the exit velocity has zero swirl. The rotor rotates at 3000 rpm and has a mean radius of 1.53 m. The fluid (steam) at inlet has static properties of 100 bar and 700 °C, at outlet has static pressure of 1 bar. The designed constant axial velocity is 150 m/s.

i) Calculate the absolute and relative flow angles upstream and downstream of the rotor.

[25%]

ii) Assuming an isentropic process and repeating stages, estimate the number of stages required.

[25%]

iii) For a 60 MW power output, calculate the required blade height at turbine entry. Does the calculated value seem reasonable? If not, what could be done to adjust it? You may assume the working fluid is a perfect gas here.

[25%]

continued

Q2

(a) Using the fundamental laws demonstrate why the total binding energy increases in exothermic nuclear reaction.

[15%]

(b) Explain why no moderator is used in the LMFBR.

[5%]

(c) Reactor core consists of a very dilute homogenous mixture of natural uranium and heavy water moderator with the following properties:
Average number of neutrons per fission from ^{235}U, $\nu = 2.5$
Fission cross section for ^{235}U, $\sigma_f(^{235}\text{U}) = 580$ b
Absorption cross section for ^{235}U, $\sigma_a(^{235}\text{U}) = 700$ b
Absorption cross section for ^{238}U, $\sigma_a(^{238}\text{U}) = 2.7$ b
Absorption cross section for D_2O , $\sigma_a(D_2O) = 1.3 \times 10^{-3}$ b
Abundance (a/o) in natural uranium for ^{235}U, $\gamma = 0.0072$
Abundance (a/o) in natural uranium for ^{238}U can be taken as $1 - \gamma = 0.9928$
Taking into account that the moderator-to-fuel ratio by mass is 50 calculate the ratio of number of molecules of D_2O to the number of all atoms of uranium in the mixture.

[15%]

(d) Assuming the resonance escape probability, p, is 0.8 and neglecting effect of leakage calculate the reproduction constant. Comment if neutron chain reaction is sustained.

[20%]

(e) Using the equivalence between mass and energy on an atomic scale 1 u = 931.5 MeV and the data in Table Q2 calculate the Q-value and the Q-value per nucleon of the following fusion reaction:
$^2\text{H} + {}^2\text{H} \rightarrow \text{n} + {}^3\text{He}$

[10%]

isotope	symbol	Atomic number (Z)	Atomic mass (u)
proton	P	0	1.007277
neutron	N	0	1.008665
protium	^{1}H	1	1.007825
deuterium	^{2}H	1	2.014102
tritium	^{3}H	1	3.016049
helium	^{3}He	2	3.016029

Table Q2

(f) Sketch typical steam expansions through the turbines of PWR nuclear power cycle on a T-s diagram. You can refer to the steam-water T-s diagram in Fig. Q2. Comment on the steam conditions at the inlets and at the exits from the steam turbines.

[20%]

(g) Compare the typical electric power outputs from AGR and PWR nuclear power cycles. Explain the differences in the steam outputs from the nuclear reactor steam generators in these two types of power cycle, that produce the different power output levels.

[15%]

continued

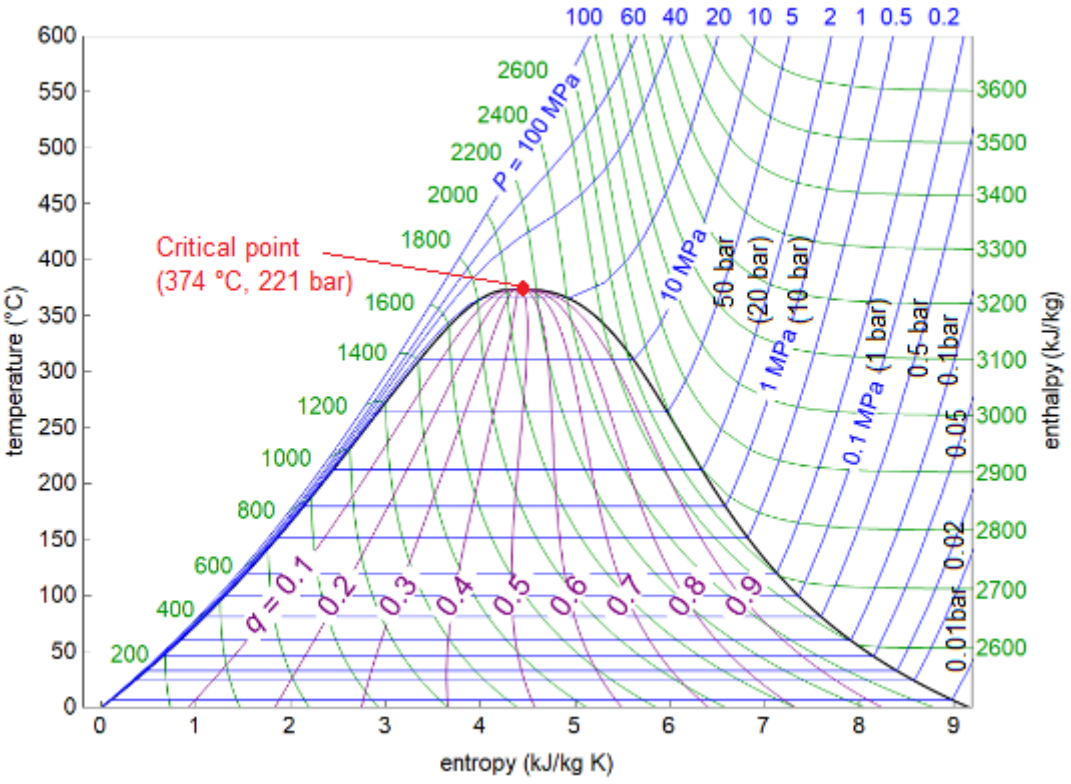


Fig. Q2

continued

Q3

(a) A convergent-divergent nozzle is supplied with air from a reservoir which holds a fixed stagnation pressure. The flow is adiabatic and quasi-one-dimensional. Describe the flow regimes which occur inside the nozzle as the back pressure is reduced relative to the upstream stagnation pressure. Sketch the variation of the static pressure along the length of the nozzle and label the key features if any.

[15%]

(b) In a certain type of space rocket engine, a convergent-divergent nozzle is connected to a reservoir of gas at a stagnation pressure of 1.5 bar. The nozzle is designed to provide isentropic acceleration of the gas to supersonic speed at exit conditions. At the design condition, the exit static pressure is 0.06 bar. The diameter of the throat is 0.2 m. Throughout this question, assume a perfect gas with the properties of air.

i) Calculate the exit Mach number and the area of the nozzle exit at this condition.

[20%]

ii) Show that a shock will appear in the divergent part of the nozzle when launching at sea level. The reservoir conditions are considered unchanged.

[25%]

iii) At a certain altitude, a normal plane shock is found at a location where the local nozzle cross sectional area is two times the throat cross sectional area. Calculate the ambient pressure at this altitude.

[20%]

iv) Calculate the momentum thrust $\dot{m}u$ at the altitude determined in iii), where $\dot{m}$ and u are the mass flow rate and the velocity at the nozzle exit plane, respectively. Is it possible to increase the momentum thrust without changing the reservoir condition?

[20%]

END